MATH 1210 Worksheet 5 Summer 2023

Suggested exercises, more exercises can be found in the textbook or on Dr. R.S.D. Thomas' webpage for archived material:

`http://home.cc.umanitoba.ca/~thomas/Courses`

Remark: The square of a matrix A is defined by $A^2 = A \cdot A$. Similarly, we have $A^3 = A \cdot A^2 = A \cdot A \cdot A$, and $A^4 = A \cdot A^3 = A \cdot A \cdot A \cdot A$. Recursively we define $A^n = A \cdot A^{n-1}$.

1. Show that
$$\begin{pmatrix} 0 & 3 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}^3$$
is the zero matrix.

2. Prove by induction that
$$\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}^n = \begin{pmatrix} 2n+1 & -4n \\ n & 1-2n \end{pmatrix}$$
for any $n \geq 1$.

3. Consider the vectors $\mathbf{u} = \langle 1, -2, -3 \rangle$, $\mathbf{v} = \langle 2, 2, -1 \rangle$ and $\mathbf{w} = 3\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$, the points $P(2, 3, -1)$ and $Q(1, 0, 4)$.
   (a) Is the vector joining the points P and Q perpendicular to $\mathbf{u} + \mathbf{v}$?
   (b) Find a unit vector parallel to $2\mathbf{u} - \mathbf{w}$.
   (c) Find a vector of length 2 in the opposite direction to $\mathbf{u}$
   (d) Find a vector of length 3 perpendicular to $\mathbf{v}$ and to the x-axis.

4. Consider the vectors $\mathbf{u} = \langle m, 2m, -1 \rangle$ and $\mathbf{v} = \langle 2m, 1, 3 \rangle$ and $\mathbf{w} = \langle 1, 0, m \rangle$ where m is a real number.
   (a) For which values of m are the vectors $\mathbf{u}$ and $\mathbf{v}$ perpendicular?
   (b) For which values of m are the vectors $\mathbf{u}$ and $\mathbf{w}$ perpendicular?

5. Find an equation of the plane containing the points $P_1(2, 3, 4)$, $P_2(1, -1, 0)$ and $P_3(3, 1, 1)$.

6. Consider the plane $x + y - 4z = -3$.
   (a) Which one of the following three points lies on the plane?
   $$P_1(3, 2, 2) \qquad P_2(5, 4, 3) \qquad P_3(2, -1, 7)$$
   (b) Is the line
   $$\begin{cases} x = 2 + 3t \\ y = -1 + 5t \\ z = 1 + 2t \end{cases}$$
   contained in the plane?

7. Find an equation parallel to the xz-plane and containing the point $P(2, 2, 5)$.

8. Find an equation of the plane through the point $P(1,-1,2)$ and intersecting the plane $2x-3y+z=5$ along the line

$$\begin{cases} x = 1+t \\ y = 2+t \\ z = 9+t \end{cases}$$

9. Find symmetric equations of the line

$$\begin{cases} x = 2+3t \\ y = 3-2t \\ z = 1+2t \end{cases}$$

Then find two planes that intersect along the line.

10. For which values of p and q is the following matrix in reduced row echelon form?

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 3-q & 1 \\ 0 & 0 & p^2+qp-q-1 \end{pmatrix}$$

11. What is the intersection of the planes $-2x+5y+3z=5$ and $-2y+3z=-1$? Describe it geometrically.

12. Use Gauss Jordan elimination to solve the system

$$\begin{aligned} x+2y-z &= 3 \\ 5x+4y+z &= 3 \\ 2x+y &= -1 \\ 2x+2y-z &= 1 \end{aligned}$$

13. Find basic solutions of the homogeneous system

$$\begin{aligned} x_1-x_2+3x_3-x_4 &= 0 \\ x_1+x_2+x_3+3x_4 &= 0 \\ 2x_1+x_2+3x_3+4x_4 &= 0 \\ 3x_1+x_2+5x_3+5x_4 &= 0. \end{aligned}$$

14. Consider the system

$$\begin{aligned} -x_1-x_2+2x_3-4x_4 &= 0 \\ 2x_1+x_2-4x_3+5x_4 &= 0 \\ 4x_1+x_2-8x_3+7x_4 &= 0. \end{aligned}$$

   (a) Without solving the system, show that it cannot have a unique solution.
   (b) Use Gauss-Jordan elimination to solve the system.
   (c) Find basic solutions of the system.